

Data Quality Dimensions

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A Classification Framework for Data Quality Dimensions

1. **Accuracy**, correctness, validity, and precision focus on the adherence to a given reality of interest.
2. **Completeness**, pertinence, and relevance refer to the capability of representing all and only the relevant aspects of the reality of interest.
3. **Redundancy**, minimality, compactness, and conciseness refer to the capability of representing the aspects of the reality of interest with the minimal use of informative resources.
4. **Readability**, comprehensibility, clarity, and simplicity refer to ease of understanding and fruition of information by users.

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5. **Accessibility** and availability are related to the ability of the user to access information from his or her culture, physical status/functions, and technologies available.
6. **Consistency**, cohesion, and coherence refer to the capability of the information to comply without contradictions to all properties of the reality of interest, as specified in terms of integrity constraints, data edits, business rules, and other formalisms.
7. **Usefulness**, related to the advantage the user gains from the use of information.
8. **Trust**, including believability, reliability, and reputation, catching how much information derives from an authoritative source. The trust cluster encompasses also issues related to security.

Accuracy Cluster

- Accuracy is defined as the closeness between a data value v and a data value v_0 , considered as the correct representation of the real-life phenomenon that the data value v aims to represent.
 - As an example if the name of a person is John, the value $v = \text{John}$ is correct, while the value $v = \text{Jhn}$ is incorrect.
- The world around us changes, and what we have called in the above definition “the real-life phenomenon that the data value v aims to represent” reflects such changes. We call temporal accuracy such type of accuracy, in contrast to structural accuracy (or, simply, accuracy), that characterizes the accuracy of data as observed in a specific time frame, where the data value can be considered as stable and unchanged.
- In the following, we will consider first structural accuracy and later temporal accuracy.

Structural Accuracy

- Two kinds of (structural) accuracy can be identified, namely, a syntactic accuracy and a semantic accuracy.
- **Syntactic accuracy** is the closeness of a value v to the elements of the corresponding definition domain D .
- In syntactic accuracy, we are not interested in comparing v with the true value v_0 ; rather, we are interested in checking whether v is any one of the values in D , whatever it is.
- So, if $v = \text{Jack}$, even if $v_0 = \text{John}$, v is considered syntactically correct, as Jack is an admissible value in the domain of persons' names.
- Syntactic accuracy is measured by means of functions, called comparison functions, that evaluate the distance between v and the values in D .
- Edit distance is a simple example of a comparison function, taking into account the minimum number of character insertions, deletions, and replacements to convert a string s to a string s_0 . More complex comparison functions exist, for instance, taking into account similar sounds or character transpositions.

Example of Syntactic Accuracy

- Let us consider the relation Movies illustrated below.
- The value Rman Holiday in movie 3 for Title is syntactically inaccurate, since it does not correspond to any title of a movie. Roman Holiday is the closest movie name to Rman Holiday; indeed, the edit distance between Rman Holiday and Roman Holiday is equal to 1 and simply corresponds to the insertion of the char o in the string Rman Holidays. Since 1 is the edit distance, the measure of syntactic accuracy is 1.

Id	Title	Director	Year	#Remakes	LastRemakeYear
1	Casablanca	Weir	1942	3	1940
2	Dead Poets Society	Curtiz	1989	0	NULL
3	Rman Holiday	Wylder	1953	0	NULL
4	Sabrina	NULL	1964	0	1985

Structural Accuracy: Semantic Accuracy

- Semantic accuracy is the closeness of the value v to the true value v_0 .
- Let us consider again the relation Movies. The exchange of directors' names in tuples 1 and 2 is an example of a semantic accuracy error: indeed, for movie 1, a director named Curtiz would be admissible, and thus it is syntactically correct.
- Nevertheless, Curtiz is not the director of Casablanca; therefore a semantic accuracy error occurs.

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Syntactic vs. Semantic Accuracy

- The above examples clearly show the difference between syntactic and semantic accuracy.
- Note that while it is reasonable to measure syntactic accuracy using a distance function, semantic accuracy is measured better with a <yes, no> or a <correct, not correct> domain.
- In contrast with what happens for syntactic accuracy, in order to measure the semantic accuracy of a value v , the corresponding true value has to be known, or, else, it should be possible, considering additional knowledge, to deduce whether the value v is or is not the true value.

Checking Semantic Accuracy

- In your opinion, in a more general context, how can we check semantic accuracy? Any idea?

Checking Semantic Accuracy

- In a more general context, a technique for checking semantic accuracy consists of looking for the same data in different data sources and finding the correct data by comparisons.
- This latter approach also requires the solution of the object **identification problem**, i.e., the problem of understanding whether two tuples refer to the same real-world entity or not.

Granularity of Accuracy

- The accuracy discussed above is referred to a single value of a relation attribute.
- In practical cases, coarser accuracy definitions and metrics may be applied. As an example, it is possible to calculate:
 - the accuracy of an attribute called attribute (or column) accuracy,
 - The accuracy of a relation (relation accuracy),
 - or the accuracy of a whole database (database accuracy).

Time-Related Accuracy Dimensions

- A relevant aspect of data is their change and update in time. From temporal point of view, data can be classified into three categories: stable, long-term-changing, and frequently changing data.
- The principal time- related dimensions proposed for characterizing the above three types of data are currency, volatility, and timeliness.

Currency

- Currency concerns how promptly data are updated with respect to changes occurring in the real world.
- As an example, the attribute #Remakes of movie 4 has low currency because a remake of movie 4 has been done, but this information did not result in an increased value for the number of remakes.
- Similarly, if the residential address of a person is updated, i.e., it corresponds to the address where the person lives, then the currency is high.

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Volatility

- Volatility characterizes the frequency with which data vary in time.
- For instance, stable data such as birth dates have volatility equal to 0, as they do not vary at all.
- Conversely, stock quotes, a kind of frequently changing data, have a high degree of volatility due to the fact that they remain valid for very short time intervals.
- Volatility is defined as the length of time data remains valid.

Timeliness

- Timeliness expresses how current the data are for the task at hand.
- The timeliness dimension is motivated by the fact that it is possible to have current data that are actually useless because they are late for a specific usage.
- For instance, the timetable for university courses can be current by containing the most recent data, but it is not timely if it is available only after the start of the classes.

Completeness Cluster

- Completeness can be generically defined as “the extent to which data are of sufficient breadth, depth, and scope for the task at hand”.
- Three types of completeness can be identified.
 - Schema completeness is defined as the degree to which concepts and their properties are not missing from the schema.
 - Column completeness is defined as a measure of the missing values for a specific property or column in a table.
 - Population completeness evaluates missing values with respect to a reference population.

Consistency Cluster

- The consistency dimension captures the violation of semantic rules defined over (a set of) data items, where items can be tuples of relational tables or records in a file.
- With reference to relational theory, integrity constraints are an instantiation of such semantic rules.

Integrity Constraints

- It is possible to distinguish two main categories of integrity constraints, namely, intra-relation constraints and inter-relation constraints.
- **Intra-relation integrity constraints** can regard single attributes (also called domain constraints) or multiple attributes of a relation.
- Let us consider an Employee relational schema, with the attributes Name, Surname, Age, WorkingYears, and Salary.
 - An example of the domain constraint defined on the schema is “Age is included between 0 and 120.”
 - An example of a multiple attribute integrity constraint is “If WorkingYears is less than 3, then Salary could not be more than 25.000 Euros per year.”

Integrity Constraints

- **Inter-relation integrity constraints** involve attributes of more than one relation.
- As an example, consider the Movies relational instance. Let us consider another relation, OscarAwards, specifying the Oscar awards won by each movie and including an attribute Year corresponding to the year when the award was assigned.
- An example of interrelation constraint states that “Year of the Movies relation must be equal to Year of OscarAwards.”

Data Edits

- Where data are not relational, consistency rules can still be defined.
- As an example, in the statistical field, data coming from census questionnaires have a structure corresponding to the questionnaire schema.
- The semantic rules are thus defined over such a structure in a way very similar to relational constraints.
- Such rules, called edits, are less powerful than integrity constraints because they do not rely on a data model like the relational one. Nevertheless, data editing has been done extensively in the national statistical agencies since the 1950s and has revealed a fruitful and effective area of application.
- Data editing is defined as the task of detecting inconsistencies by formulating rules that must be respected by every correct set of answers. Such rules are expressed as edits, which denote error conditions.

Data Edits

- As an example, an inconsistent answer to a questionnaire can be to declare

- `marital status = "married", age = "5 years old"`
The rule to detect this kind of errors could be the following:

- The rule can be put in the form of an edit, which expresses the error condition, namely,
`if marital status is married, age must not be less than 14.`

`marital status = married  $\wedge$  age < 14`

Trade-Offs Between Dimensions

- Data quality dimensions are not independent, i.e., correlations exist between them.
- If one dimension is considered more important than the others for a specific application, then the choice of favoring it may imply negative consequences for the other ones.
- First, trade-offs may need to be made between timeliness and any one of the three dimensions: accuracy, completeness, and consistency.
- Indeed, having accurate (or complete or consistent) data may need checks and activities that require time, and thus timeliness is negatively affected. Conversely, having timely data may cause lower accuracy (or completeness or consistency).

Trade-Offs Between Dimensions

- A typical situation in which timeliness can be preferred to accuracy, completeness, or consistency is given by most Web applications: as the time constraints are often very stringent for Web data, it is possible that such data are deficient with respect to other quality dimensions.
- For instance, a list of courses published on a university Web site must be timely though there could be accuracy or consistency errors and some fields specifying courses could be missing.
- Conversely, when considering an administrative application, accuracy, consistency, and completeness requirements are more stringent than timeliness, and therefore delays are mostly admitted in dimensions other than timeliness.

Trade-Offs Between Dimensions

- Another significant case of trade-off is between consistency and completeness.
- Here the question is “Is it better to have less but consistent data, i.e., poor completeness, or to have more but inconsistent data?”
- This choice is again very domain specific.
- As an example, statistical data analysis typically requires a significant and representative amount of data in order to perform the analysis; in this case, the approach is to favor completeness, tolerating inconsistencies or adopting techniques to solve them.